

QuickBite — Complete Database Entity Relationship Diagram

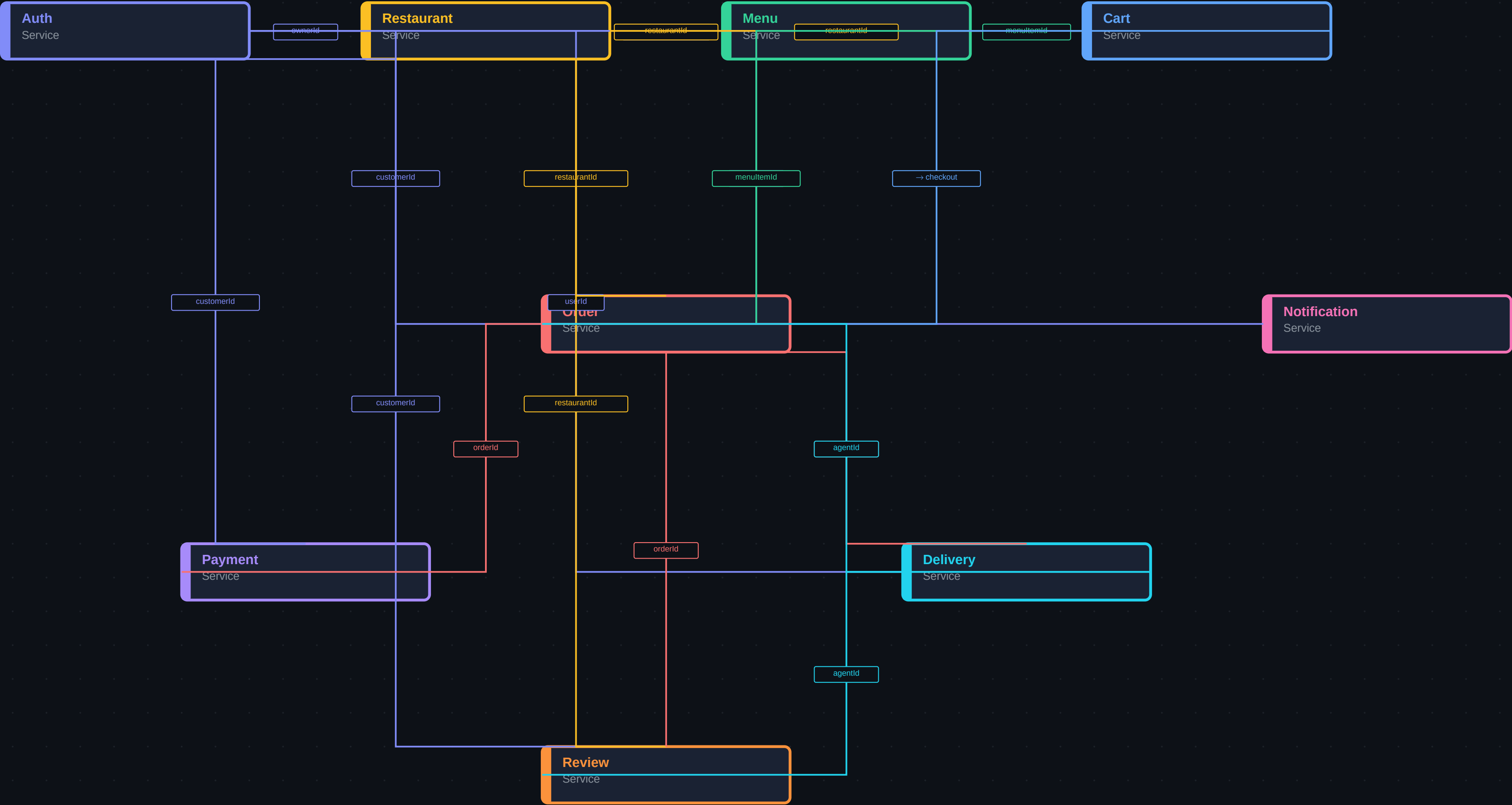


LEGEND

- PK** Primary Key – unique row identifier
- FK** Foreign Key – references another entity
- UK** Unique Constraint
- Data type (shown right-aligned)
- 1:N relationship (same service)
- Cross-service FK reference

QuickBite — Cross-Service Reference Map

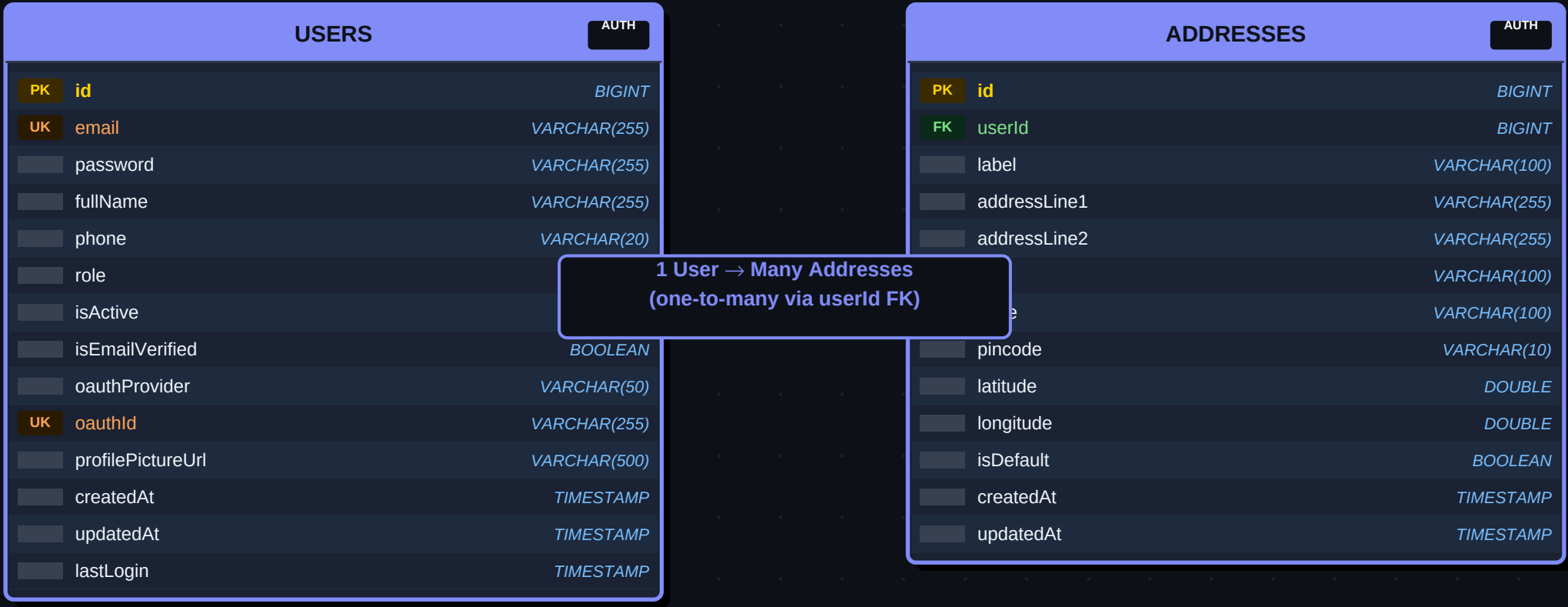
How entities reference each other across microservice boundaries



Auth Service

Port 8081 · PostgreSQL · auth_db

Manages authentication, user profiles, OAuth2 (Google/GitHub), JWT tokens, and delivery addresses.



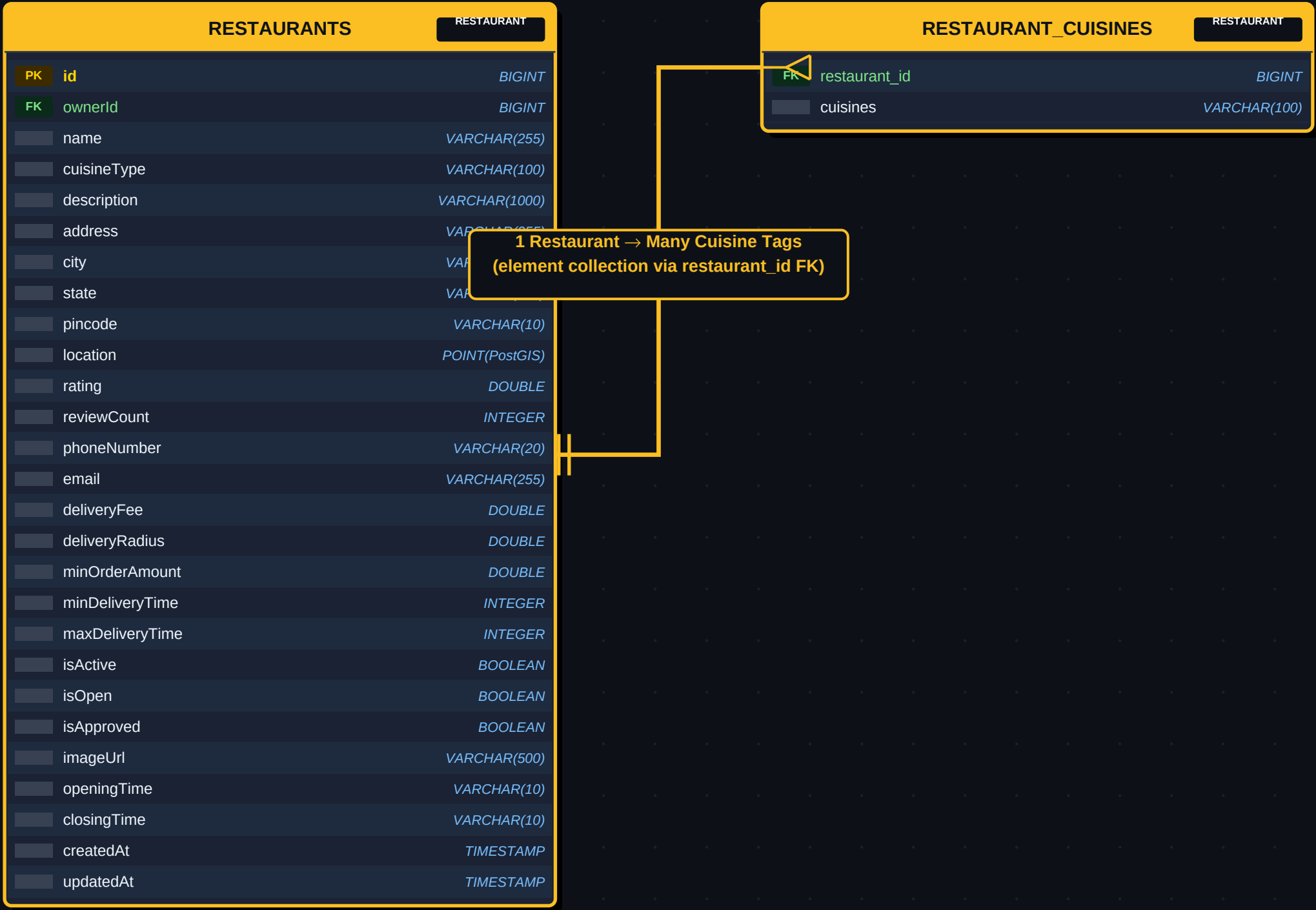
CROSS-SERVICE REFERENCES:

users.id → restaurants.ownerId, carts.customerId, orders.customerId, payments.customerId, delivery_agents.userId, reviews.customerId, notifications.userId

Restaurant Service

Port 8082 · PostgreSQL + PostGIS · restaurant_db

Manages restaurant profiles, location (PostGIS POINT), operating hours, and cuisine tags.



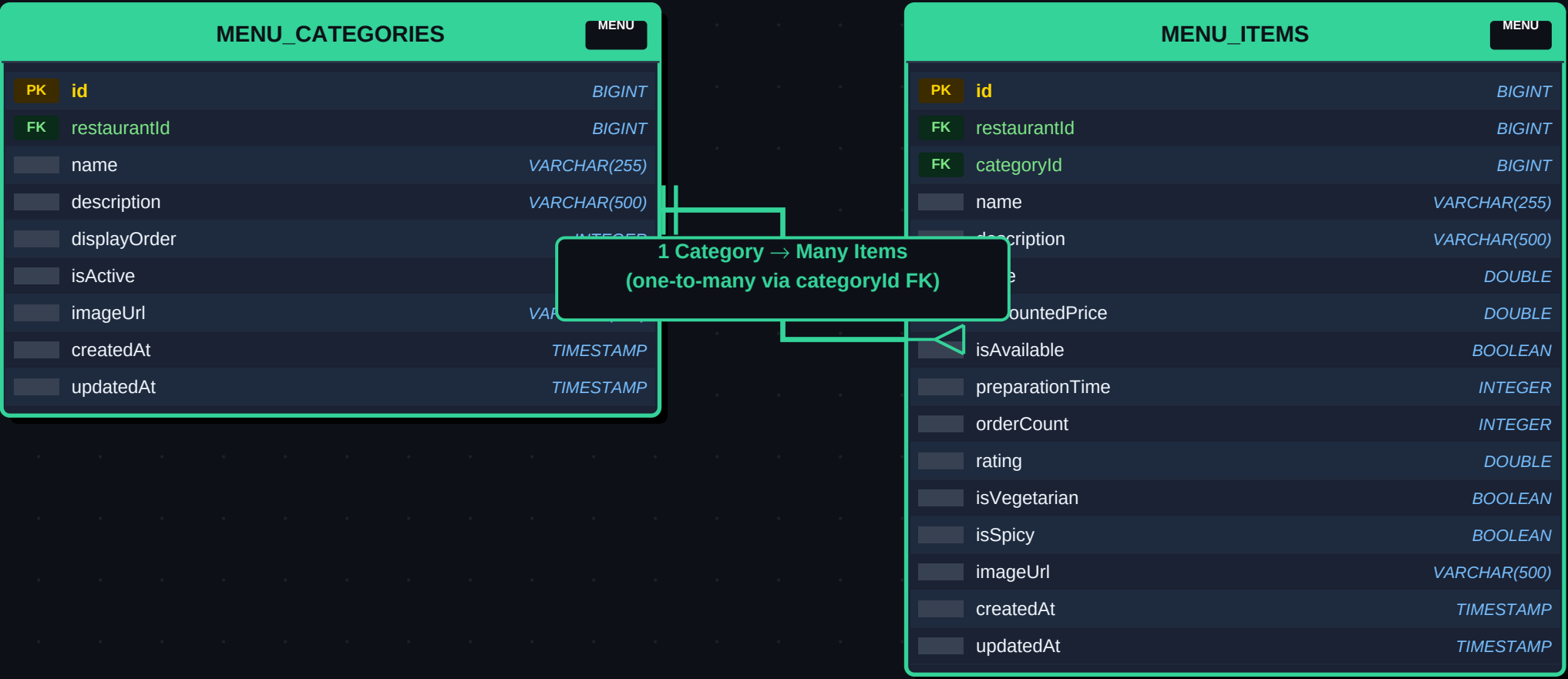
CROSS-SERVICE REFERENCES:

restaurants.id → menu_categories.restaurantId, menu_items.restaurantId, shopping_carts.restaurantId, orders.restaurantId, reviews.restaurantId

Menu Service

Port 8083 · PostgreSQL · menu_db

Manages hierarchical menu: categories group items. Cloudinary used for image storage.



CROSS-SERVICE REFERENCES:

menu_items.id → cart_items.menuItemId, order_items.menuItemId

Cart Service

Port 8084 · PostgreSQL · cart_db

Manages per-customer shopping cart. One active cart per customer at a time.



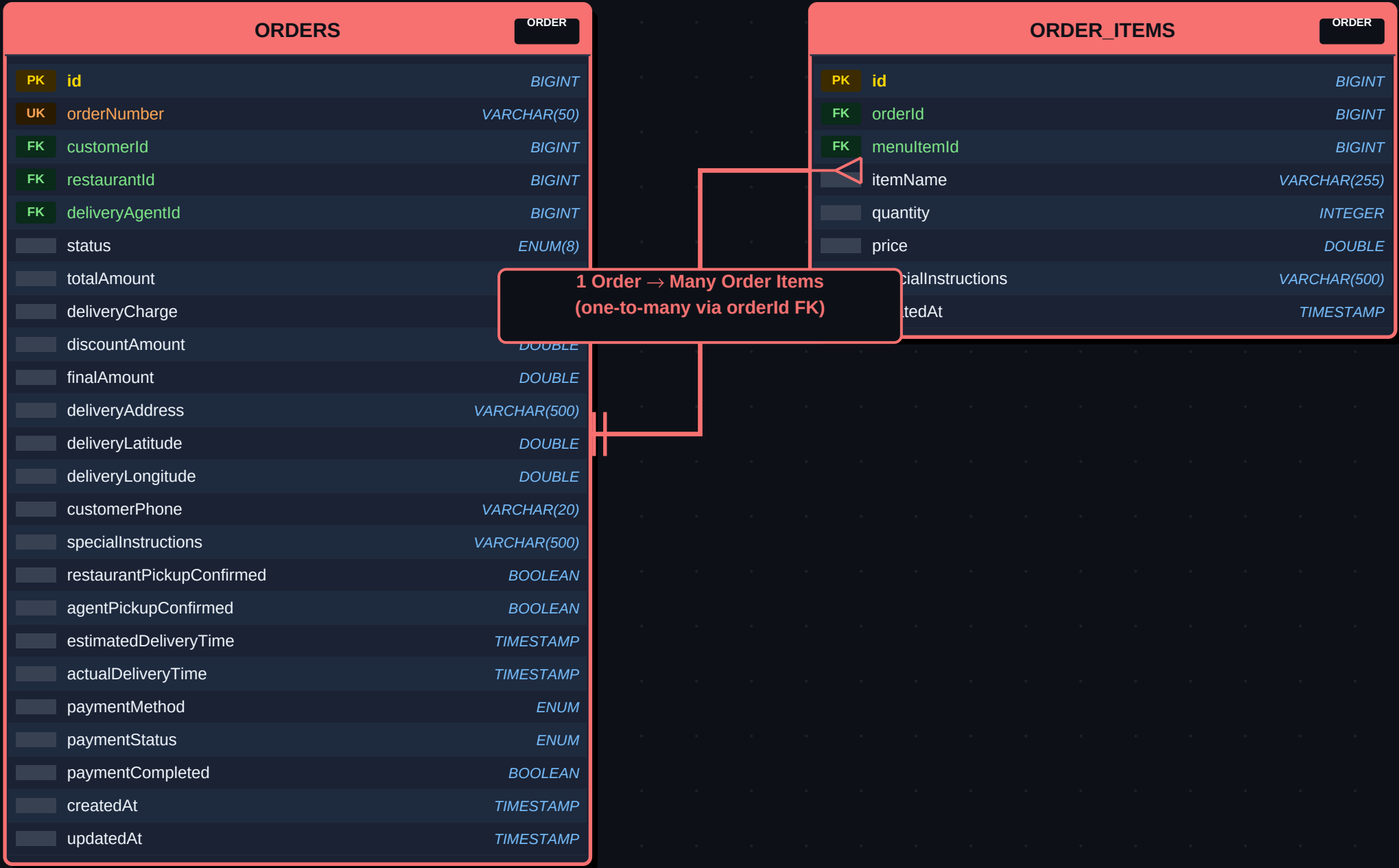
CROSS-SERVICE REFERENCES:

shopping_carts.customerId ← users.id | shopping_carts.restaurantId ← restaurants.id | cart_items.menuItemId ← menu_items.id

Order Service

Port 8085 · PostgreSQL · order_db

Tracks full order lifecycle: PENDING → CONFIRMED → PREPARING → READY → PICKED_UP → DELIVERED.



CROSS-SERVICE REFERENCES:

orders.customerId ← users.id | orders.restaurantId ← restaurants.id | orders.deliveryAgentId ← delivery_agents.id | order_items.menuItemId ← menu_items.id

Payment Service

Port 8086 · PostgreSQL · payment_db

Handles Razorpay gateway, wallet top-up/debit, COD, and promo code discounts.

PAYMENTS		PAYMENT
PK	id	BIGINT
UK	transactionId	VARCHAR(100)
FK	orderId	BIGINT
FK	customerId	BIGINT
	amount	DOUBLE
	status	ENUM(4)
	paymentMethod	ENUM
	razorpayPaymentId	VARCHAR(255)
	razorpayOrderId	VARCHAR(255)
	razorpaySignature	VARCHAR(500)
	failureReason	VARCHAR(500)
	currency	VARCHAR(3)
	refundedAt	TIMESTAMP
	createdAt	TIMESTAMP
	updatedAt	TIMESTAMP

WALLETS		PAYMENT
PK	id	BIGINT
FK	customerId	BIGINT
	balance	DOUBLE
	createdAt	TIMESTAMP
	updatedAt	TIMESTAMP

1 Wallet → Many Transactions
(one-to-many via walletId FK)

WALLET_TRANSACTIONS		PAYMENT
PK	id	BIGINT
FK	walletId	BIGINT
	customerId	BIGINT
	transactionType	ENUM
	amount	DECIMAL(10,2)
	balanceBefore	DECIMAL(10,2)
	balanceAfter	DECIMAL(10,2)
	description	VARCHAR(500)
FK	referenceId	BIGINT
	referenceType	VARCHAR(50)
	createdAt	TIMESTAMP

PROMO_CODES		PAYMENT
PK	id	BIGINT
UK	code	VARCHAR(50)
	description	VARCHAR(500)
	discountType	ENUM
	discountValue	DOUBLE
	minOrderAmount	DOUBLE
	maxDiscountAmount	DOUBLE
	validFrom	TIMESTAMP
	validUntil	TIMESTAMP
	usageLimit	INTEGER
	usageCount	INTEGER
	isActive	BOOLEAN
	createdAt	TIMESTAMP
	updatedAt	TIMESTAMP

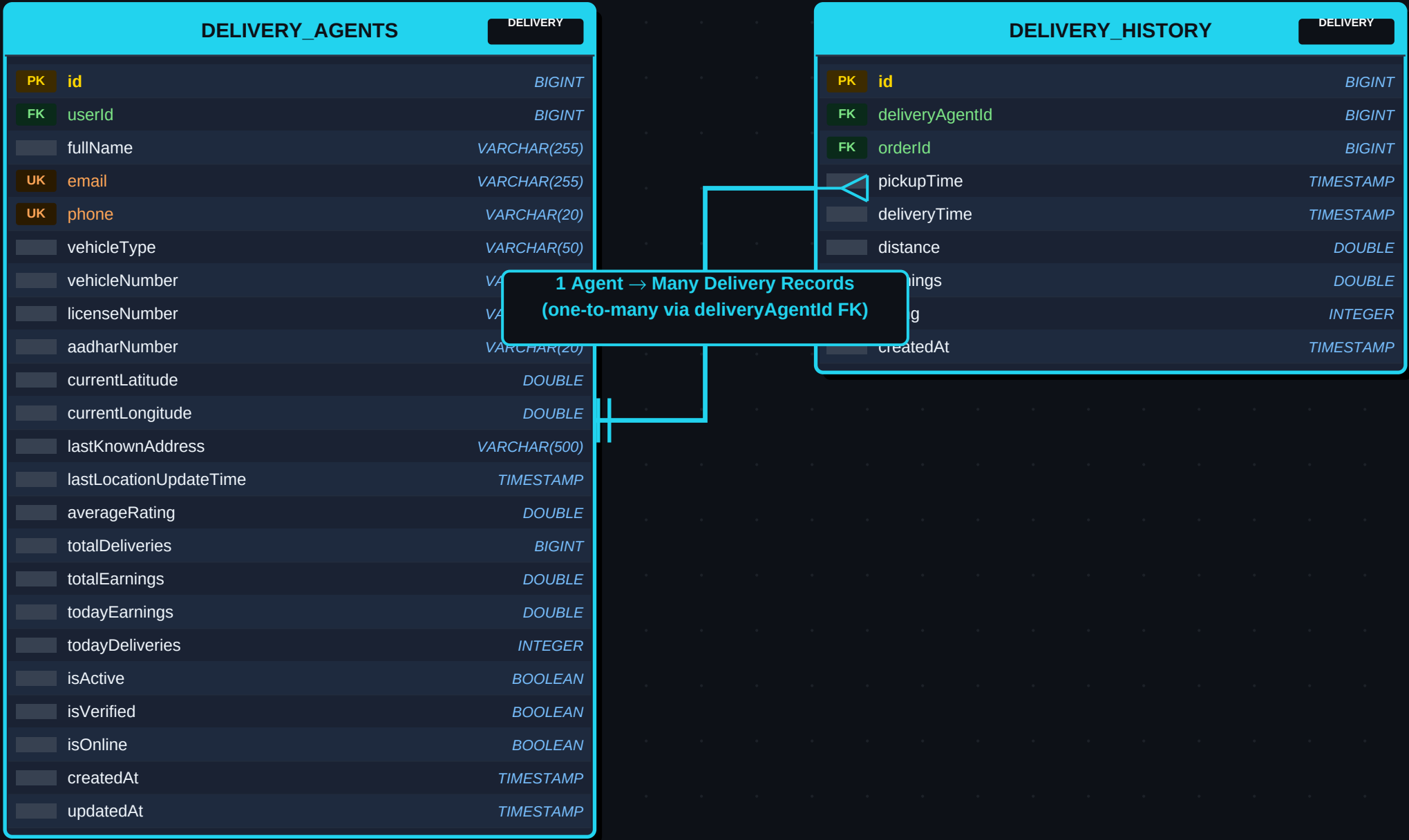
CROSS-SERVICE REFERENCES:

payments.orderId ← orders.id | payments.customerId / wallets.customerId / wallet_transactions.customerId ← users.id

Delivery Service

Port 8087 · PostgreSQL + PostGIS · delivery_db

Tracks delivery agent location in real-time (PostGIS), earnings, and delivery records.



CROSS-SERVICE REFERENCES:

delivery_agents.userId ← users.id | delivery_history.orderId ← orders.id

Review Service

Port 8088 · PostgreSQL · review_db

Stores post-delivery reviews for restaurant AND delivery agent. Triggers rating aggregation.

REVIEWS			REVIEW
PK	id	BIGINT	
FK	orderId	BIGINT	
FK	customerId	BIGINT	
FK	restaurantId	BIGINT	
FK	deliveryAgentId	BIGINT	
	restaurantRating	INTEGER	
	restaurantReview	VARCHAR(500)	
	deliveryRating	INTEGER	
	deliveryReview	VARCHAR(500)	
	isAnonymous	BOOLEAN	
	createdAt	TIMESTAMP	
	updatedAt	TIMESTAMP	

CROSS-SERVICE REFERENCES:

reviews.orderId ← orders.id | reviews.customerId ← users.id | reviews.restaurantId ← restaurants.id | reviews.deliveryAgentId ← delivery_agents.id

Notification Service

Port 8089 · PostgreSQL · notification_db

Event-driven notifications (WebSocket/SSE). Triggered by order status changes.

NOTIFICATIONS			NOTIFICATION
PK	id	BIGINT	
FK	userId	BIGINT	
	title	VARCHAR(255)	
	message	VARCHAR(1000)	
	type	VARCHAR(50)	
	eventType	VARCHAR(100)	
	referenceId	BIGINT	
	isRead	BOOLEAN	
	createdAt	TIMESTAMP	
	readAt	TIMESTAMP	

CROSS-SERVICE REFERENCES:

notifications.userId ← users.id | notifications.referenceId → order/payment/delivery entity